A FLAT MAP OF LOCALLY NOETHERIAN SPACES FROM A GORENSTEIN SOURCE IS GORENSTEIN WITH GORENSTEIN TARGET LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6653-6660 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6662-6667, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Gorenstein implies Cohen-Macaulay, but the lemma is strictly stronger than L16052 and is proved from a different algebra input.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The translation of the Gorenstein property for spaces into Noetherian local rings (lemma-gorenstein, same chapter), the parallel argument in lemma-composition-gorenstein, and the dualizing-complexes algebra result lemma-flat-under-gorenstein.

Source reference. Original lemma 0BJL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6667–6667. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E16: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6612–6627. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E15: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6538–6550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a flat morphism of locally Noetherian algebraic spaces over S. If X is Gorenstein, then f is Gorenstein and $\mathcal{O}_{Y,f(\overline{x})}$ is Gorenstein for all $x \in |X|$.*

Proof. After translating into algebra using Lemma 0E15 (compare with the proof of Lemma 0E16) this follows from Dualizing Complexes, Lemma 0BJL. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.